

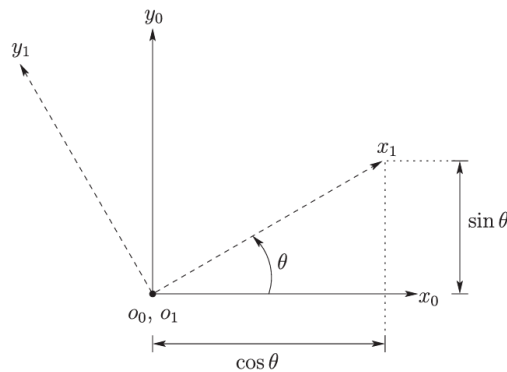
Dynamic Orientation Estimation with Sensor Fusion Of 9 - Axis Inertial Measurement Unit (IMU) Sensors

EE 2120: Electrical Measurements and Instrumentation

1. REPRESENTING ROTATIONS

In order to represent the relative position and orientation of one rigid body with respect to another, we will rigidly attach coordinate frames to each body, and then specify the geometric relationships between these coordinate frames.

ROTATIONS IN 2D



NOTATION: Consider, $\mathbf{x}_n^b$, Here, b represents the frame with respect to which $\mathbf{x}_n$ is defined (in this case, frame **b**).

e.g.- $\mathbf{P}^2$ indicates point P is defined with respect to the frame 2.

Two coordinate frames shown, with frame $o_1x_1y_1$ obtained by rotating frame $o_0x_0y_0$ by an angle θ .

Specify the coordinate vectors for the axes of frame $o_1x_1y_1$ with respect to coordinate frame $o_0x_0y_0$:

$$R_1^0 = \begin{pmatrix} | & | \\ x_1^0 & y_1^0 \\ | & | \end{pmatrix}$$

Where, x_1^0 and y_1^0 are the coordinates of the unit vectors x_1 and y_1 in the $o_0x_0y_0$ coordinate frame

The R_1^0 is called the Rotation Matrix & for the 2D case, it can be calculated as:

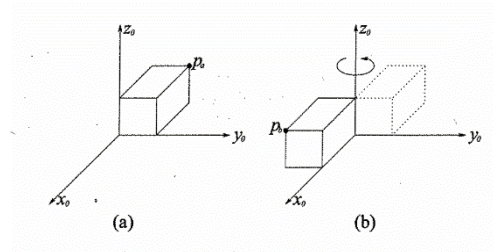
$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

NOTE: A rotation matrix may represent 3 operations, let $p = [p_1 \ p_2 \ p_3]$ represent a point in 3D space or a vector.

- a. Changing the frame (change coordinates / change basis) of a vector or another frame

E.g: $p^a = R_b^a p^b$ will change the basis of vector p from frame b to frame a

- b. Representing a rotation in the current frame.



Notice the rotation of point p on the box. Suppose, there is a frame (x_1, y_1, z_1) : *frame 1*, attached to the box, in (a) (x_1, y_1, z_1) coincides with (x_0, y_0, z_0) : *frame 0*. In (b), when the box is rotated, (x_1, y_1, z_1) rotates with the body. For (b), We can calculate R_1^0 by writing basis vectors of frame 1, w.r.t frame 0. Then, $p^0 = R_1^0 p^1$ will transform the basis from frame 1 to frame 0. p^0 represents the rotated vector p according to *frame 0*.

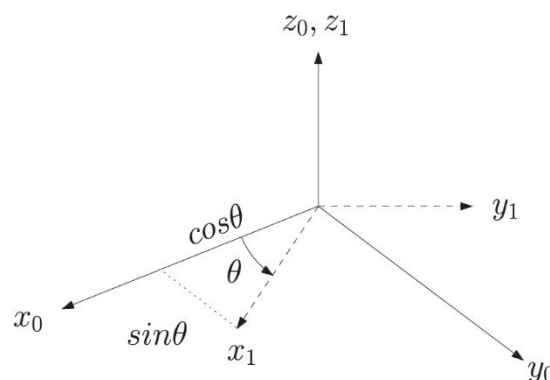
- c. Representing the orientation of one frame relative to another.

$$R_1^0 = \begin{pmatrix} | & | \\ x_1^0 & y_1^0 \\ | & | \end{pmatrix}$$

Example: Verify, $p^0 = R_1^0 p^1$ with $p = [1 \ 0]$ and $R_1^0 = \begin{bmatrix} \cos(45) & -\sin(45) \\ \sin(45) & \cos(45) \end{bmatrix}$

ROTATIONS IN 3D

Rotation about z_0 by an angle θ .



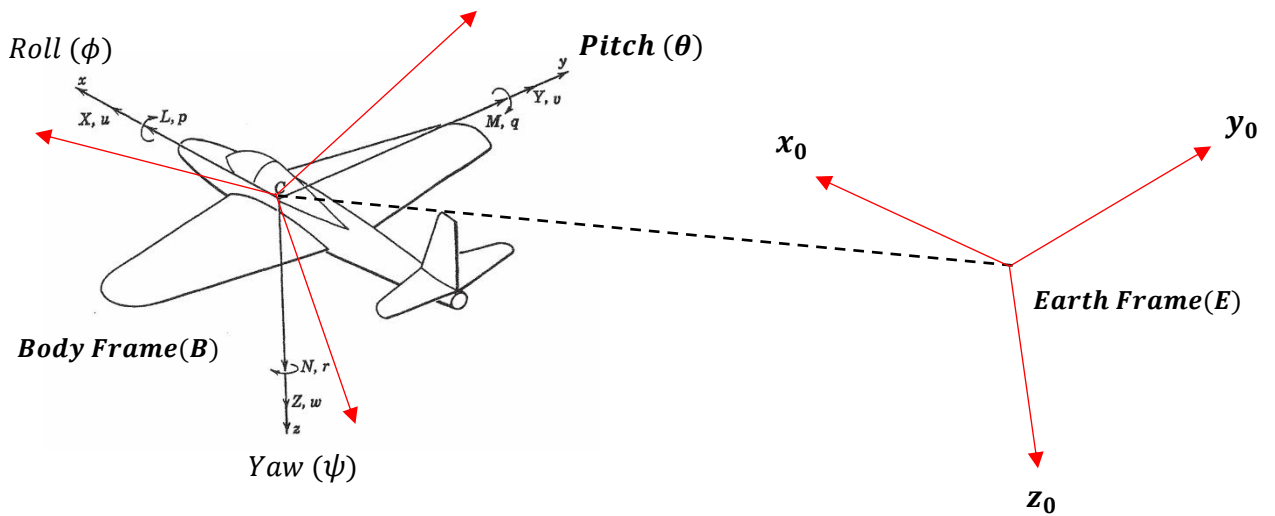
$$R_1^0 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} = R_{z,\theta}$$

Similarly, Rotations about x and y axes,

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$R_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

2. EULER ANGLES



The term Euler Angles refers to the angles of rotation (ψ, θ, ϕ) needed to go from one coordinate system to another using the specific sequence of rotations.

The most common sequence is:

$$\text{Yaw}(\psi) \rightarrow \text{Pitch}(\theta) \rightarrow \text{Roll}(\phi)$$

$$Z \rightarrow Y \rightarrow X$$

$$p^{Body} = [R_x(\phi)R_y(\theta)R_z(\psi)]p^{Earth}$$

$$p^{body} = R_E^B p^{Earth}$$

$$\therefore p^{Earth} = [R_x(\phi)R_y(\theta)R_z(\psi)]^{-1} p^{Body} = R_B^E p^{Body}$$

Gravity in the body frame can be written as,

$$g^B = R_E^B g^E = R_E^B \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

Since $g^E = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$, what we need is the 3rd row of $R_x(\phi)R_y(\theta)R_z(\psi)$ to calculate g^B

$$\therefore g^B = \begin{bmatrix} g \sin(\theta) \\ -g \cos(\theta) \sin(\phi) \\ -g \cos(\theta) \cos(\phi) \end{bmatrix}$$

3. WHY SENSOR FUSION?

- Overcome limitations of individual sensors (Noise, uncertainty, Bias, Drift, Sampling rates)
- Estimate quantities that are not directly measurable or too expensive to measure.

4. SENSORS AND SENSOR MODELS

- 3 – Axis Accelerometer: Measures linear acceleration along 3 principal axes x, y, z of the sensor (m/s^2)
- 3 – Axis Gyroscope: Measures angular velocity along 3 principal axes x, y, z of the sensor (rad/s^2)
- 3 – Axis Magnetometer: Measures the magnetic field along 3 principal axes of the sensor (T)

ACCELEROMETER

Measures accelerations along each of the body frame axes.

$$\hat{a} = a^B = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \underbrace{\frac{d\hat{v}}{dt}}_{\text{Linear Acceleration}} + \underbrace{\hat{\omega}_b \times \hat{v}}_{\text{Rotational translation}} - \underbrace{R_0^1 \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}}_{\text{gravity}} + \underbrace{\beta(t)}_{\text{Bias}} + \underbrace{\eta(t)}_{\text{Noise}}$$

If the body is at rest,

$$a^B = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} g \sin(\theta) \\ -g \cos(\theta) \sin(\phi) \\ -g \cos(\theta) \cos(\phi) \end{bmatrix} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Bias} \\ | \end{bmatrix}$$

GYROSCOPE

Measures accelerations along each of the body frame axes.

$$\omega^B = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} | \\ \omega_{true} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Bias} \\ | \end{bmatrix} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix}$$

Transforming body angular rates from Gyroscope to Euler Rates in the fixed frame,

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Process model

Euler-rate kinematics: The gyro rates p, q, r (rad/s , $body\ x, y, z$) are mapped to Euler rates. Note that, accelerometer carries no heading information:

$$\begin{aligned} \dot{\phi} &= p + \sin(\phi)\tan(\theta)q + \cos(\phi)\tan(\theta)r \\ \dot{\theta} &= \cos(\phi)q - \sin(\phi)r \end{aligned}$$

State vector:

$$x = [\phi \quad b_\phi \quad \theta \quad b_\theta]^T$$

Where, $\phi = roll$, $\theta = pitch$, and b_ϕ, b_θ, b_ψ are *bias* states modeled as random-walk constants

Discrete - time process model.

$$\begin{aligned} \phi_k &= \phi_{k-1} + \Delta t(\dot{\phi} - b_\phi) \\ b_{\phi,k} &= b_{\phi,k-1} \\ \theta_k &= \theta_{k-1} + \Delta t(\dot{\theta} - b_\theta) \\ b_{\theta,k} &= b_{\theta,k-1} \end{aligned}$$

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}, \quad w \sim N(0, Q)$$

$$A = \begin{bmatrix} 1 & -\Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \Delta t & 0 \\ 0 & 0 \\ 0 & \Delta t \\ 0 & 0 \end{bmatrix}, U = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \end{bmatrix}, Q = 0.01I_4$$

Measurement model

Roll and pitch are directly provided by the accelerometer readings and are treated as measurements in the state – space model.

$$\begin{aligned} \phi_{acc} &= \text{atan2}\left(a_y, \sqrt{a_x^2 + a_z^2}\right) \\ \theta_{acc} &= \text{atan2}\left(-a_x, \sqrt{a_y^2 + a_z^2}\right) \end{aligned}$$

$$z_k = Cx_k + v_k, \quad v \sim N(0, R)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, z = \begin{bmatrix} \phi_{acc} \\ \theta_{acc} \end{bmatrix}, R = \begin{bmatrix} \sigma_\phi & 0 \\ 0 & \sigma_\theta \end{bmatrix}$$

Kalman filter equations

Predict.

$$\begin{aligned}\hat{x}^- &= A \hat{x} + Bu \\ P^- &= A P A^T + Q\end{aligned}$$

Update.

$$\begin{aligned}\tilde{y} &= z - C \hat{x}^- \\ S &= C P^- C^T + R \\ K &= P^- C^T S^{-1} \\ \hat{x} &= \hat{x}^- + K \tilde{y} \\ P &= (I - K C) P^-\end{aligned}$$

5. Extending the State Observer for estimation of complete orientation (Including yaw)

MAGNETOMETER

Measures the local magnetic field along each of the body frame axes (μT). The Earth's magnetic field expressed in the body frame is the fixed-frame field rotated into the body frame, corrupted by sensor distortions:

$$m^B = \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \underbrace{A(R_E^B m_E)}_{\text{Rotated Field and Soft Iron effect}} + \underbrace{\begin{bmatrix} | \\ b_{hi} \\ | \end{bmatrix}}_{\text{Hard iron effect}} + \begin{bmatrix} | \\ \text{Noise} \\ | \end{bmatrix}$$

Where,

m_E = Earth's field in the fixed (navigation) frame

R_E^B = rotation from fixed to body frame

A = soft iron distortions / scale factors / axis misalignments

b_{hi} = hard - iron offset (assumed to rotate with the sensor)

Calibration

Before the field can be used for heading, the hard-iron and soft-iron errors must be removed (one-time calibration). b_{hi} and A are found by rotating the sensor through all orientations and fitting an ellipsoid using convex optimization.

$$m_{cal} = \begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = A^{-1}(m_{meas} - b_{hi})$$

$$\hat{m} = \frac{m}{\|m_{cal}\|} \text{ to normalize}$$

Tilt Compensation (Correction Using Roll and Pitch)

The calibrated field still lies in the tilted body frame. To extract heading it must be de-rotated into the horizontal plane using the *roll* (ϕ) and *pitch* (θ) estimates from the filter — NOT from the magnetometer:

$$\begin{aligned} m_x^h &= m'_x \cos \theta + m'_y \sin \phi \sin \theta + m'_z \cos \phi \sin \theta \\ m_y^h &= m'_y \cos \phi - m'_z \sin \phi \end{aligned}$$

Heading (Measurement)

$$\psi_{mag} = \text{atan2}(m_y^h, m_x^h) + \delta_{decl}$$

where δ_{decl} is the local magnetic declination (include only if true north is required; omit for relative heading).

Notes:

- Roll and pitch enter only as inputs to the de-rotation; the magnetometer never corrects **ϕ or θ** .
- The innovation/ error ψ_{mag} must be wrapped to the shortest arc in $(-\pi, \pi]$ before the update.

$$v_\psi = \psi_{mag} - \hat{\psi}^- = \text{atan2}(\sin v_\psi, \cos v_\psi)$$

- For further details of the calibration procedure including proofs and derivations, refer the attached pdf.

Euler rate kinematics

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin(\phi)\tan(\theta) & \cos(\phi)\tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \frac{\sin(\phi)}{\cos(\theta)} & \frac{\cos(\phi)}{\cos(\theta)} \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

In scalar form,

$$\begin{aligned} \dot{\psi} &= \frac{\sin(\phi)}{\cos(\theta)} q + \frac{\cos(\phi)}{\cos(\theta)} r \\ \dot{\phi} &= p + \sin(\phi)\tan(\theta) q + \cos(\phi)\tan(\theta) r \\ \dot{\theta} &= \cos(\phi) q - \sin(\phi) r \end{aligned}$$

State vector:

$$x = [\phi \quad b_\phi \quad \theta \quad b_\theta \quad \psi \quad b_\psi]^T$$

Where, ϕ = *roll*, θ = *pitch*, ψ = *yaw* and b_ϕ, b_θ, b_ψ are *bias* states modeled as random-walk constants

Discrete - time process model.

$$\begin{aligned}
 \phi_k &= \phi_{k-1} + \Delta t(\dot{\phi} - b_\phi) \\
 b_{\phi,k} &= b_{\phi,k-1} \\
 \theta_k &= \theta_{k-1} + \Delta t(\dot{\theta} - b_\theta) \\
 b_{\theta,k} &= b_{\theta,k-1} \\
 \psi_k &= \psi_{k-1} + \Delta t(\dot{\psi} - b_\psi) \\
 b_{\psi,k} &= b_{\psi,k-1}
 \end{aligned}$$

$$\begin{aligned}
 x_k &= Ax_{k-1} + Bu_{k-1} + w_{k-1}, \quad w \sim N(0, Q) \\
 A &= \begin{bmatrix} 1 & -\Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -\Delta t & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -\Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \Delta t & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \Delta t \\ 0 & 0 & 0 \end{bmatrix}, u = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, Q = 0.01I_4
 \end{aligned}$$

Measurement model

Roll and pitch are directly provided by the accelerometer readings and are treated as measurements in the state – space model.

$$\begin{aligned}
 \phi_{acc} &= \text{atan2}\left(a_y, \sqrt{a_x^2 + a_z^2}\right) \\
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 \end{aligned}$$

$$\begin{aligned}
 m_{cal} &= \begin{bmatrix} m'_x \\ m'_y \\ m'_z \end{bmatrix} = A^{-1}(m_{meas} - b_{hi}) \\
 \hat{m} &= \frac{m_{cal}}{\|m_{cal}\|}
 \end{aligned}$$

Tilt removal using roll and pitch

$$\begin{aligned}
 m_x^h &= m'_x \cos \theta + m'_y \sin \phi \sin \theta + m'_z \cos \phi \sin \theta \\
 m_y^h &= m'_y \cos \phi - m'_z \sin \phi
 \end{aligned}$$

Heading (Measurement)

$$\psi_{mag} = \text{atan2}(m_y^h, m_x^h) + \delta_{decl}$$

Measurement

$$\begin{aligned}
 z_k &= Cx_k + v_k, \quad v \sim N(0, R) \\
 z &= \begin{bmatrix} \phi_{acc} \\ \theta_{acc} \\ \psi_{mag} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, R = \begin{bmatrix} \sigma_\phi & 0 & 0 \\ 0 & \sigma_\theta & 0 \\ 0 & 0 & \sigma_\psi \end{bmatrix}
 \end{aligned}$$